

# INTELLIGENT HUMAN RESOURCE AND WORKFORCE PLATFORM

DBMS Project Review 2 | Literature Review  
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# Problem Background

Employee data is scattered across multiple disconnected systems and spreadsheets, making it hard to access or maintain.

HR tasks such as leave requests, approvals, and onboarding are handled manually, leading to delays and errors.

Managers lack a single unified view of team workload, attendance, and performance metrics. Traditional HR systems mainly store data and provide limited intelligent insights for decision-making.

Problem Statement: Organizations need an integrated, intelligent HR platform that centralizes data, automates workflows, and delivers actionable analytics.

# Project Objectives

1. Centralize Employee Data
2. Build a unified PostgreSQL database to store all employee records, profiles, contracts, and organizational data in one secure, accessible location.
3. Manage Attendance, Leave, Tasks & Performance
4. Provide modules for tracking daily attendance, processing leave requests, assigning tasks, and monitoring individual performance metrics in real time.

# Project Objectives (Cont.)

1. Automate workforce approvals and workflows — reduce manual HR processing by enabling rule-based and intelligent automation for leave, task assignments, and approval chains.
2. Provide intelligent analytics — generate data-driven insights on attendance, performance, and workload trends to support proactive workforce planning.
3. Support HR and manager decisions while keeping human control — offer recommendations and dashboards that assist decision-makers without replacing human judgment in critical HR matters.

# Literature Review Method

Sources: Scopus, ScienceDirect, Springer, and Google Scholar were searched for relevant literature.

Keywords Used: AI in HRM, Intelligent HR, Machine Learning in HRM, AI recruitment, employee analytics, workforce management.

Selection Criteria: Papers were screened by relevance, recency, and research quality. Only peer-reviewed articles and systematic reviews were included.

Comparison Approach: Selected papers were compared by research method, key findings, and identified limitations to synthesize insights for the project.

# Selected Papers

Jatobá et al. (2023) — AI-based HRM

Systematic review of 61 Scopus articles. AI supports recruitment, performance, engagement and training. Limitation: narrow scope per study.

Pan & Froese (2023) — AI and HRM

Systematic review of 184 articles. Research is broad but fragmented across HR functions. Limitation: lacks integration view.

Sakib & Islam (2026) — ML in HRM

Systematic review + bibliometric analysis. ML supports prediction and automation in HR. Limitation: limited real-world deployment evidence.

2025 — AI Recruitment Review

Systematic review of 49 studies. Efficiency improves with AI-driven recruitment. Limitation: bias and transparency remain key concerns.

2026 — AI and DEI Review

Systematic review of 43 studies. AI can standardize HR processes but may reproduce existing bias. Limitation: DEI outcomes vary by context.

All 5 papers were sourced from Scopus, ScienceDirect, Springer and Google Scholar and compared by method, findings and limitations.

# Detailed Analysis

Jatobá et al. (2023)

- Method: Systematic review of Scopus literature
- Evidence Base: 61 peer-reviewed articles
- Findings: AI supports recruitment, performance management, employee engagement, and training programs
- Limitation: Focuses on individual HR functions; lacks integrated system perspective

Pan & Froese (2023)

- Method: Systematic literature review
- Evidence Base: 184 articles across HRM and AI domains
- Findings: AI adoption in HRM is growing across multiple functions and industries
- Limitation: Research is broad but fragmented; no unified framework proposed

Sakib & Islam (2026)

- Method: Systematic review combined with bibliometric analysis
- Evidence Base: Multi-source academic literature
- Findings: ML effectively supports HR prediction tasks (turnover, performance) and workflow automation
- Limitation: Limited focus on integrated database-centered platforms; data quality and bias remain concerns

Key Takeaway

- All three papers confirm AI/ML adds value to HR processes
- Each paper focuses on separate HR functions rather than a unified platform
- None address a DBMS-centered system combining employee data, workflows, and analytics in one solution

# Critical Comparison

HR Area: Recruitment, Performance, Training, Retention, Engagement

Method: Systematic reviews (Scopus, ScienceDirect); bibliometric analysis

Strength: AI/ML improves automation, prediction & decision support across HR

Limitation: Most studies focus on single HR functions; limited integration scope

Integration: Few papers address one unified DB + workflow + analytics platform

Similarities: All three key papers confirm AI/ML enhances HR automation and predictive capability. Each validates data-driven approaches over manual processes.

Differences: Jatobá focuses on broad AI-HRM functions; Pan & Froese highlight fragmentation across 184 studies; Sakib & Islam emphasize ML prediction and automation depth.

Key Insight: Papers agree AI/ML improves HR — but most target individual functions. A single integrated DBMS-centered platform with workflows and analytics remains largely unaddressed.

# Synthesis & Challenges

Common Findings across literature:

- Automation of repetitive HR tasks improves efficiency
- Predictive analytics supports workforce planning and retention
- AI-driven decision support enhances performance management

Common Challenges identified:

- Algorithmic bias in recruitment and evaluation
- Privacy and data protection concerns
- Lack of transparency in AI decisions (black-box models)
- Poor data quality undermines model reliability
- Need for human oversight to validate AI outputs

Key Takeaway: Intelligent HR systems should assist and empower HR professionals and managers — not blindly replace human judgment in workforce decisions.

# Research Gap

Existing Research: Studies focus separately on recruitment, performance, training, or retention — resulting in fragmentation with no unified view.

Gap: Limited integration of employee database, workforce workflows, and intelligent analytics within a single DBMS-centered platform.

Our Contribution: A unified platform combining centralized employee data, workflow management, intelligent analytics, and human-in-the-loop decision support — built on DBMS principles.

# Proposed System & Conclusion

System Architecture: Users → Application Layer → HR/Workforce Modules → PostgreSQL Database → Analytics → Decision Support → HR/Manager

- Literature confirms AI/ML improves automation, prediction, and decision support in HR
- Existing research is fragmented — few studies integrate database, workflows, and analytics in one platform
- Our project unifies centralized employee data, workforce workflow management, and intelligent analytics using core DBMS principles with human-in-the-loop control

Key References: Jatobá et al. (2023) · Pan & Froese (2023) · Sakib & Islam (2026) · AI Recruitment Review (2025) · AI + DEI Review (2026)

**THANK YOU**